

Mind the Gap to Trustworthy LLM Agents: A Systematic Evaluation on Constraint Satisfaction for Travel Planning

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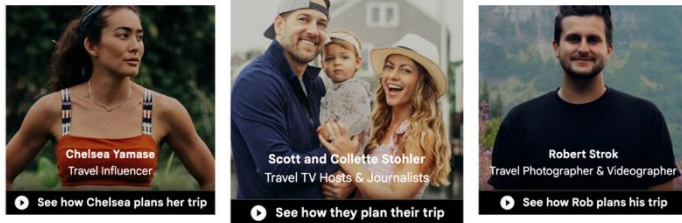
Travel Planning: A Practical Scenarios

❖ Travel Planning: The Consensus 'Killer App' and Proving Ground for Next-Gen Agentic AI.



Tongyi DeepResearch

Trip.com



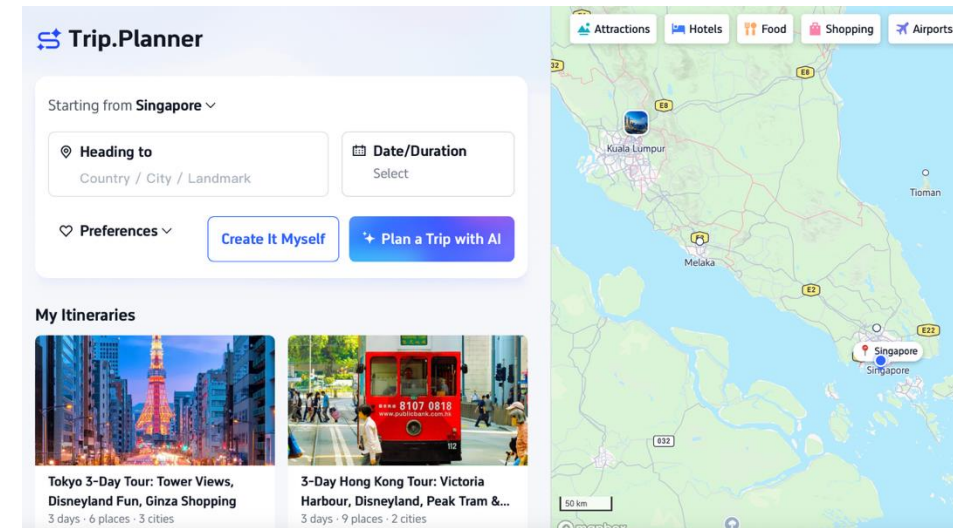
Go anywhere
with Google

Find great prices on flights and hotels,
discover must-see sights, and plan your
ideal trip – all starting with a simple search.

Plan your trip



Detailed and realistic travel
itineraries

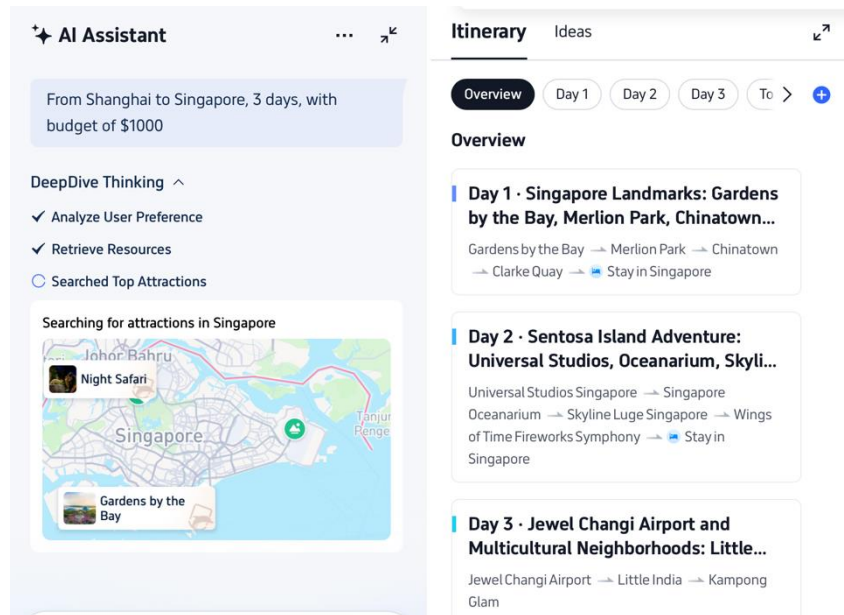


<https://travel.google>
<https://tongyi-agent.github.io/showcase/>
<https://sg.trip.com/webapp/tripmap/tripplanner>

Can We Trust the Travel Agent?

❖ Agentic Travel Planner

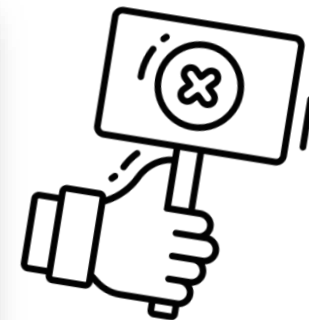
- Query Understanding
- Information Collection
- Plan Generation



Constraint Satisfaction in Travel Planning



ence plans. Comprehensive evaluations show that the current language agents are not yet capable of handling such complex planning tasks—even GPT-4 only achieves a success rate of 0.6%. Language agents struggle to stay on task, use the right tools to collect information, or keep track of multiple constraints. However, we note that the mere

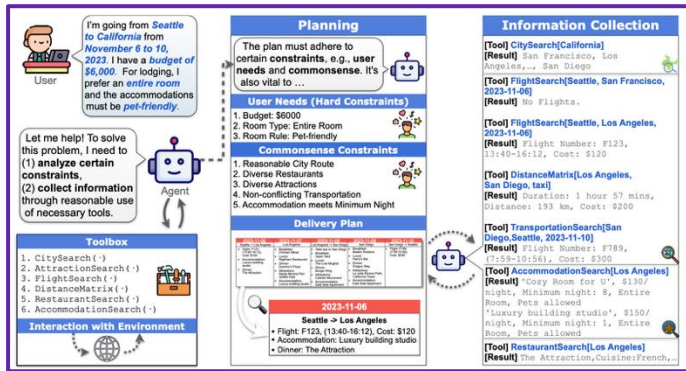


In this Work

Landscape Review

Fine-Grained Modular Decomposition

Benchmarks Evolution



Constraint Extraction

Evaluating how models identify and interpret specific requirements from natural language

Information Integration

Testing tool-calling capabilities and plan production while respecting the extracted constraints

Reasoning under Constraints

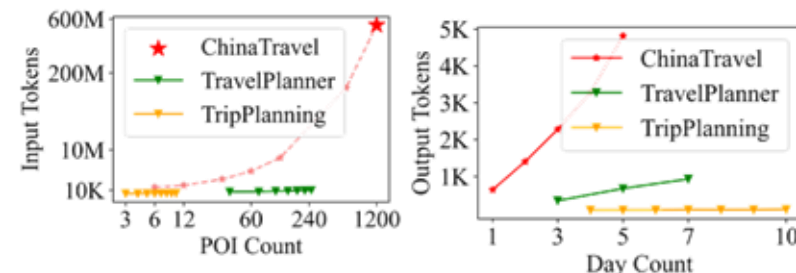
Auditing the final reasoning steps to ensure spatial-temporal feasibility and trustworthy.



DAYS	1	2	3	4	5
 DESTINATION	Welton Hotel (Room 229)	The Atrium Event Centre	The Atrium Event Centre	The Asewell	The Asewell
 EAT	The Garrison, Wanderlust Pub Ali Khan's	Lunch & Dinner Provided	The Hakman Magic Show	Julianne's, Mario Italian, The Hot Wok	The Captain's Mack, Jackson Asian, Spicing Rolls
 LEISURE	Jazz Festival Monero	Yak-Yak Comedy Club	Yak-Yak Comedy Club	The Networkers Marketing Event	The Networkers Marketing Event
 TRANSPORTATION	Transport, Hotel Cab	Public Transport, Cab	Public Transport, Cab	Public Transport, Cab	Public Transport, Cab



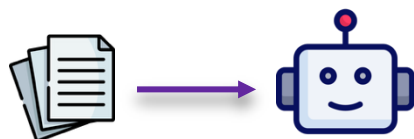
Date/Time	Locations	Event	Notes
14 January 2011 8:00 - 10:00 PM	Supper	Light to Heavy	Light Supper, Chinese Buffet, Business CAs
15 January 2011 11:00 - 12:00 PM	Hong Kong Airport Hong Kong Hotel	Arrive at Hong Kong Check out of Hong Kong	Pick up bag, Hotel Services Business Partner Services
16 January 2011 11:00 - 12:00 PM	Hong Kong Hotel Hong Kong Hotel	Heavy golfing CAs	Pick up bag, HongKong company at hotel Hong Kong Hotel, member phone
17 January 2011 11:00 - 12:00 PM	Hong Kong Restaurant	Dinner with business partner	Business Partner Services Business Partner Services
18 January 2011 09:00 - 09:40 AM 10:00 - 10:40 AM	Hong Kong Hotel	Exhibition	Business Partner Services Business Partner Services
19 January 2011 10:40 AM - 14:00 PM	Hong Kong Trade Centre	Exhibition	Foreign HR - Company Card to Hong Kong/English meeting next day Back to hotel using my own MTR or taxi
20 January 2011 11:00 - 12:00 PM	Hong Kong Restaurant	Dinner	Any place or no hotel
21 January 2011 10:00 - 10:40 AM 10:40 - 11:00 AM 11:00 - 12:00 PM	Hong Kong Hotel Business Partner Services Business Partner Services	Breakfast meeting with business partner Business Partner Services Business Partner Services	Breakfast to hotel Restaurant Business Partner Services Business Partner Services
22 January 2011 11:00 - 12:00 PM	Hong Kong	Free time, city tour	Business Partner Services Business Partner Services
23 January 2011 11:00 - 12:00 PM	Hong Kong	Breakfast meeting	Breakfast to hotel Restaurant
24 January 2011 11:00 - 12:00 PM	Hong Kong Hotel	Check out from Hong Kong	Hotel Services
25 January 2011 11:00 - 12:00 PM	Light to Supper	Light to Supper	Light Supper, Chinese Buffet, Business CAs
26 January 2011 09:00 - 09:40 AM	Supper at Airport	Arrive at Airport	Pick up bag, HongKong company at hotel Hong Kong Hotel, member phone



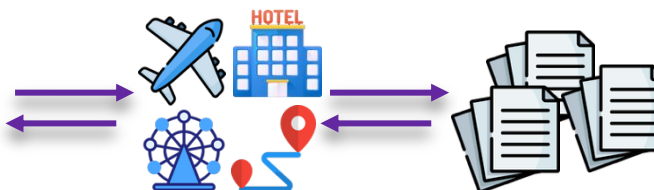
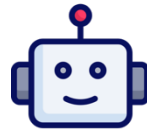
ChinaTravel[2024], TripCraft, TripTailor, RETAIL, TripTide, TripScore [2025]

Long input exceeds context limit
Long output degrades performance

Static information in context



Natural Plan [2024]



Dynamic information from tool-calling

TravelPlanner, ChinaTravel[2024], TripCraft, TripTailor, TripTide [2025]

Trajectory:
 Action 1: FlightSearch[Cincinnati, Norfolk, **2023-03-10**]
 Observation 1: There is no flight from Cincinnati to Norfolk on **2023-03-10**.
 Action 2: FlightSearch[Cincinnati, Norfolk, **2023-03-09**]
 Action 3: FlightSearch[Cincinnati, Norfolk, **2023-03-11**]
 Action 4: FlightSearch[Cincinnati, Richmond, **2023-03-10**]
 ...
 Action 8: FlightSearch[Cincinnati, Charlotte, **2023-03-10**]
 Observation 9: **I'm sorry, but as an AI language model, I am unable to execute actions or observe their outcomes.**

Trajectory:
 ... Observation 3: **The cost of your current plan is 316.0 dollars.**
 ...
 Thought 4: **I need to select a more budget-friendly accommodation** to keep the overall trip within the \$1,800 budget.
 Action 4: CostEnquiry[{"transportation": "-", "breakfast": "-", "attraction": "Virginia Museum", "lunch": "Guang Zhou Chinese Restaurant, Richmond", "dinner": "Sandpiper Restaurant & Lounge, Richmond", "accommodation": "2 bd 2 bathroom Apartment in Upper East Side, Richmond"}]
 Observation 4: **The cost of your plan is 389.0 dollars.**

Tool execution Hallucination

Templated Synthetic Data

Our budget is set at \$1,400 for this trip and we require our accommodations to be visitor-friendly. We would like to have options to dine at Indian, American, Chinese, and Italian restaurants. We also prefer not to self-drive.

Our budget is set at \$3,100. We require accommodations that are pet-friendly and we would prefer to have entire rooms to ourselves. We do not plan on self-driving for trip.

Open-World Instructions

The budget of dining is ¥1000.

Visit the Great Wall at the second day

The budget excluding flights is ¥3000.

Unseen composition with given concepts Understanding degrades

[illegible]

Constraint Understanding Gap

User Data

Templated Synthetic Data

Cost Cuisine Attraction
Transportation Duration

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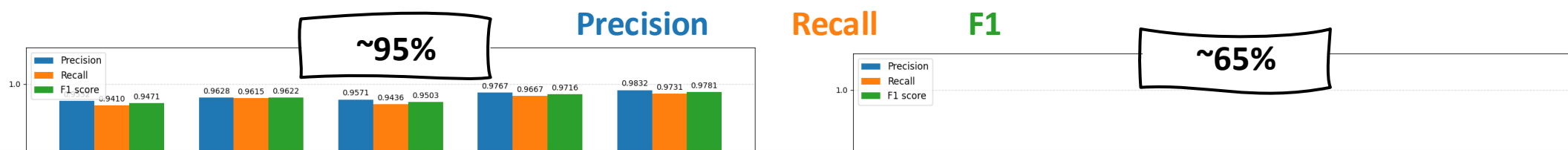
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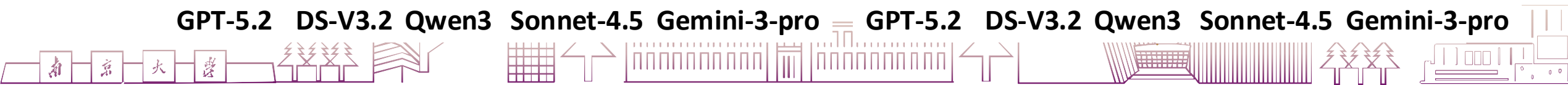
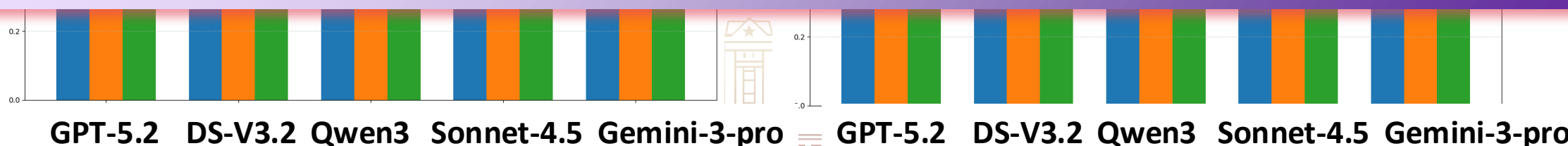
Unseen composition with given concepts Understanding degrades

Open Language Reasoning	Unseen Concept Composition
Query: I am currently in Nanjing and would like to go on a 5-day trip to Beijing with a friend. We plan to travel by high-speed train both ways and hope to try some local specialty foods. DSL Translation (GPT-4o): <pre>restaurant_type_set = set() for activity in activity(plan): if activity.type(activity) in ['breakfast', 'lunch', 'dinner']: restaurant_type_set.add(restaurant_type(activity, target_city(plan))) result = (1.4 * 1000) / len(restaurant_type_set)</pre> Query: My parents and I plan a five-day travel from Nanjing to Beijing to watch the flag-raising ceremony, and we want to stay at a hotel near Tiananmen Square. DSL Translation (GPT-4o): <pre>hotel_names_set = set() for activity in activity(plan): if activity.type(activity) == 'accommodation': hotel_names_set.add(activity.position(activity)) result = (3.1 * 1000) / len(hotel_names_set)</pre>	Query: I am traveling alone from Nanjing to Shanghai in the morning for a day trip. I plan to visit a university campus and return in the evening, making sure to catch the train back before 7 PM. DSL Translation (GPT-4o): <pre>result = True for activity in activity(plan): if activity.type(activity) == 'transport': result = False</pre> Query: My brother and I are planning to travel from Shanghai to Chongqing for 4 days. Apart from the mandatory high-speed train flights, we aim to spend no more than 1400 yuan in Chongqing. DSL Translation (GPT-4o): <pre>total_cost = 0 for activity in activity(plan): total_cost += activity.cost(activity) result = total_cost <= 1400</pre>

Q: Can LLM accurately extract constraints from natural language, especially which is unseen during the development phase?



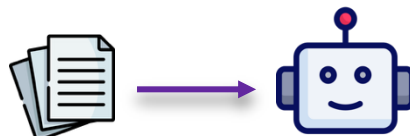
LLMs are good at following templated requirements, but struggle with the unseen compositional requirements from the real human language.



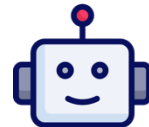
Information Integration Gap

Information Integration

State information
in context

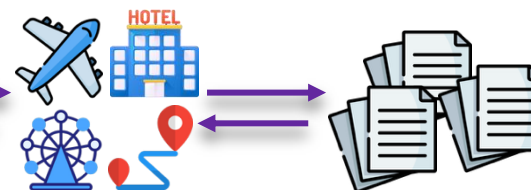


Natural Plan [2024]



Dynamic information from tool-calling

TravelPlanner, ChinaTravel[2024], TripCraft, TripTailor, TripTide [2025]

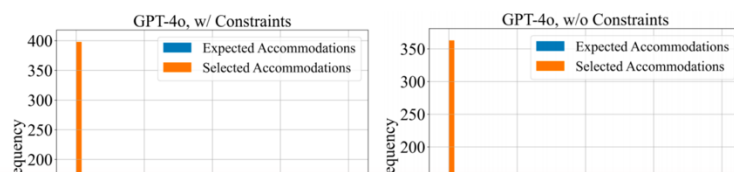


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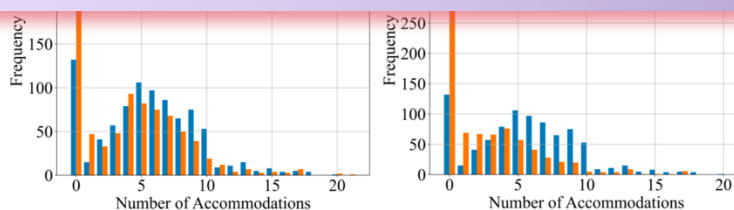
Tool execution illusion

Q: Can LLM accurately extract POIs that meet specific requirements given (ground-truth) constraints?



Setting	F1	Prec	Rec	EM
Macro/Micro	GPT-4o			

The SOTA LLMs tend to be conservative, yet they still struggle to fully satisfy all constraints, often leading to false positives despite.



w/ Const. Ann.	0.46 / 0.64	0.50 / 0.76	0.46 / 0.55	0.24
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A **low macro precision** means that the selected POIs also do not satisfy the constraints.

The **#selected POIs** << **# Groud Truth**

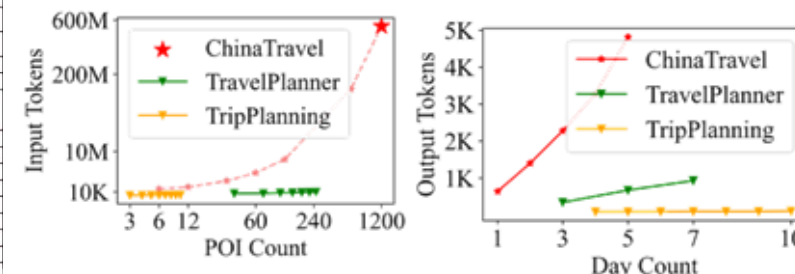


Natural Plan, TravelPlanner [2024]

DAYS	1	2	3	4	5
 DESTINATION	Welton Hotel (Room 2209)	The Atrium Event Centre	The Atrium Event Centre	The Asewell	The Asewell
 EAT	The Garrison, Winchester Pub. Ali Khan's	Lunch & Dinner Provided	The Hakman Magic Show	Johanna's, Mario Italian, The Hot Wit	The Captain's Mark Jackson Asia, Spring Boll
 LEISURE	Jazz Festival Monero	Yak-Yak Comedy Club	Yak-Yak Comedy Club	The Networkers Marketing Event	The Networkers Marketing Event
 TRANSPORTATION	Transport, Hotel Cab	Public Transport, Cab	Public Transport, Cab	Public Transport, Cab	Public Transport, Cab



Event-Level with
detailed location/time

[illegible]

Long input exceeds context limit
Long output degrades performance

Q: Can LLM produce a plan that satisfies the constraints given all the necessary information in the context?

Model	IRM	Pass Rate
Reasoning Models (LRMs), reasoning un		
OT-preview	✓	10.0

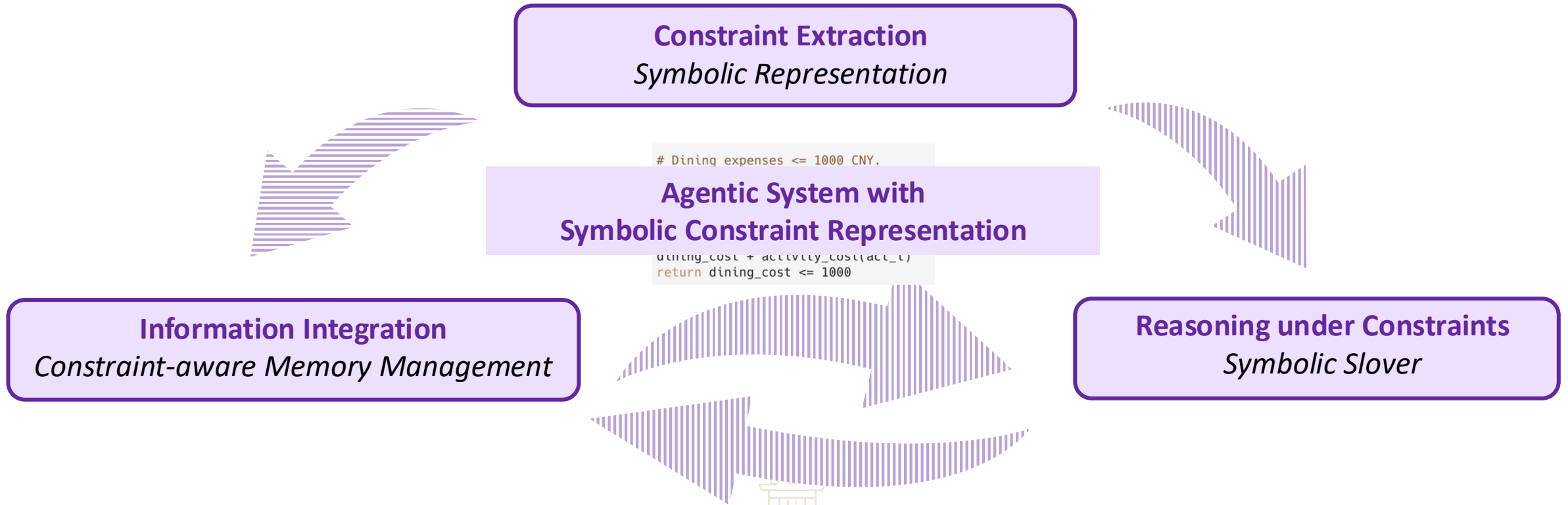
Even on **day-level tasks (TravelPlanner)**, the constraint pass rate of state-of-the-art Large Reasoning Models remains very limited.

The #selected POIs << # Groud Truth



A Promising Solution: Neuro-Symbolic Agent

- ❖ Neural networks excel at perception and understanding.
- ❖ Symbolic reasoning excels at constraint satisfaction



A Promising Solution: Neuro-Symbolic Agent

Constraint Extraction *Symbolic Representation*

Open-World Language Instructions

The budget of dining is ¥1000.

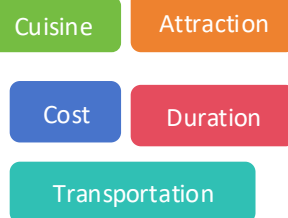
Visit the Great Wall at the second day

The budget excluding flights is ¥3000.

LLM perception



Constraint understanding



```
# Dining expenses <= 1000 CNY.  
dining_cost = 0  
for act_i in allactivities(plan):  
    typ = activity_type(act_i)  
    if typ=="breakfast" or typ=="lunch"  
    or typ=="dinner": dining_cost =  
        dining_cost + activity_cost(act_i)  
return dining_cost <= 1000
```

Unseen composition with given concepts

The program-based (symbolic) presentation supports **compositional generalization** based on basic concepts.



A Promising Solution: Neuro-Symbolic Agent

Reasoning under Constraints *Symbolic Slover*

Symbolic constraint representation

```
# Dining expenses <= 1000 CNY.  
dining_cost = 0  
for act_i in allactivities(plan):  
    typ = activity_type(act_i)  
    if typ=="breakfast" or typ=="lunch"  
or typ=="dinner": dining_cost =  
dining_cost + activity_cost(act_i)  
return dining_cost <= 1000
```

LLM generation
→
SMT code generation

Code Generator

```
# solver initiation  
solver = Optimize()  
# variable & initiation...  
total_cost = Int('total_cost')  
# ... other variables ...  
  
solver.add(total_cost == Sum(...))  
# ...more constraint assertion...  
  
# goal  
solver.minimize(total_cost)
```

Runtime Error

Z3Prover/**z3**
The Z3 Theorem Prover

Z3

315 Contributors 1 Used by 377 Discussions 12k Stars 2k Forks

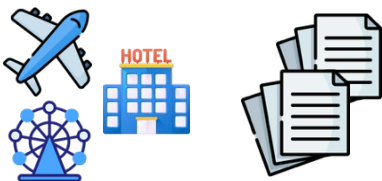
The symbolic solver supports **rigorous constraint satisfaction**.



A Promising Solution: Neuro-Symbolic Agent

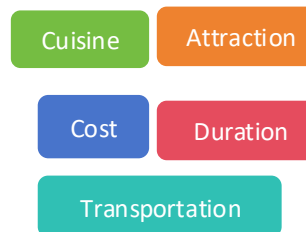
Information Integration
Constraint-aware Memory Management

Raw POI Information



LLM grounding
→
SMT code generation

Symbolic check with constraint representation



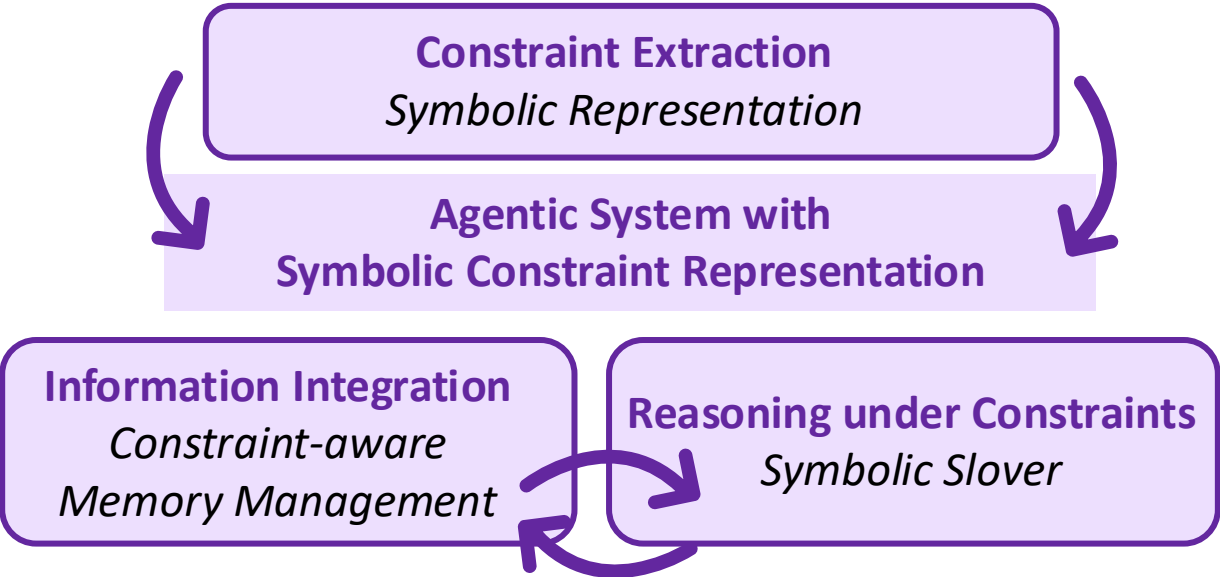
```
# Dining expenses <= 1000 CNY.  
dining_cost = 0  
for act_i in allactivities(plan):  
    typ = activity_type(act_i)  
    if typ=="breakfast" or typ=="lunch"  
    or typ=="dinner": dining_cost =  
        dining_cost + activity_cost(act_i)  
return dining_cost <= 1000
```

The symbolic check supports **accurate memory management**.



A Promising Solution: Neuro-Symbolic Agent

Neuro-Symbolic Agentic System



		DR	EPR		LPR		C-LPR FPR		DR	EPR		LPR		C-LPR FPR			
		Mic.		Mac.	Mic.	Mac.			Mic.		Mac.	Mic.	Mac.				
		Easy (#300)								Human-Val (#154)							
Act		70.4	49.9	0	64.6	30.6	0	0	-								
		97.5	70.8	0	86.8	68.6	0	0	-								
ReAct (zero-shot)		43.3	40.8	0	41.9	19.6	0	0	36.4	29.5	0.65	35.2	16.2	0.38	0		
		95.4	48.2	0	71.3	33.0	0	0	96.1	50.5	0	72.4	32.5	0	0		
ReAct (one-shot)		77.5	68.3	6.00	74.1	52.3	5.77	5.33	55.2	57.3	2.59	64.6	44.2	1.71	2.59		
		94.2	68.1	0	89.4	70.6	0	0	69.5	46.3	0	63.6	46.8	0	0		
NeSy Planning		75.3	75.3	75.3	70.4	52.6	70.4	52.6	51.9	53.2	52.5	47.0	37.6	46.5	37.0		
		75.0	73.6	64.0	73.5	63.3	61.7	60.6	45.4	50.1	45.4	40.9	29.8	38.5	27.9		
		72.3	67.0	34.0	70.4	49.6	32.6	28.3	42.8	47.4	42.2	36.2	27.2	34.4	25.3		
		32.0	31.9	31.3	29.1	21.0	28.3	21.0	25.9	25.8	24.0	22.3	12.3	20.5	11.0		
		30.3	30.3	30.3	27.6	19.6	27.6	19.6	37.6	38.2	37.6	32.7	18.8	32.2	18.8		

Pass rate: 60% on Synthetic Data
37% on Open-World Human Data

Significant improvement over pure-neural agents



Conclusion & Future Directions

❖ Travel Planning is a good domain:

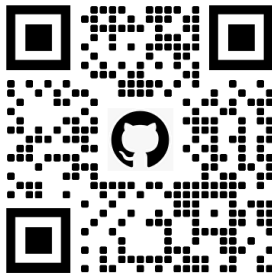
➤ a Testbed of Trustworthy Agents with both Market Value and Academic Challenges

❖ Neuro-Symbolic Agents are promising

➤ a Promising Solution towards Generalizable Trustworthy

❖ Open-world benchmarks are important:

➤ ChinaTravel: with formal verification on large-scale real human requirements



Thank you!

